Multivariate Calculus Lecture Notes
Section 16.2 – Line Integrals

I. Scalar line integrals
   a. Idea
      i. Imagine we have a curve $C$ through 3D space for which some function $f(x, y, z)$ is defined:

      ii. For example, maybe $f(x, y, z)$ represents the density at $(x, y, z)$ of some flexible tubing that is curved in the shape of $C$.
      iii. We could find the mass of the tubing in the following manner:
         1. Chop up the curve into a bunch of tiny pieces so that the density is approximately constant on each piece.
         2. Approximate the mass of each piece.
         3. Add up the masses.
         4. Take the limit as the width of the largest piece goes to 0 to get an exact value.
      iv. The result is the *line integral* (or *path integral*, or *curve integral*) of $f$ over $C$:

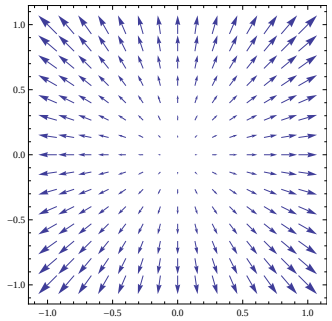
$$\int_C f(x, y, z)\, ds = \lim_{||\Delta s_i|| \to 0} \sum_{i=1}^{n} f(P_i)\Delta s_i,$$

      where
         1. $P_i$ is the point on curve piece $i$ where you are measuring the density, and
         2. $||\Delta s_i||$ is the width of the longest curve piece.
      v. Example: If $f(x, y, z) = 1$, then $\int_C ds$ is the length of the curve $C$.
   b. Calculating scalar line integrals
      i. In practice, to evaluate a line integral along a curve $C$ we need to have a parametrization $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$ of $C$.
      ii. Remember that the curve length function $s(t)$ is given by $s(t) = \displaystyle\int_a^t \|\mathbf{r}'(u)\|\, du$.
      iii. This means that, by the second part of the Fundamental Theorem of Calculus, $\dfrac{ds}{dt} = \|\mathbf{r}'(t)\|$, and so $ds = ||\mathbf{r}'(t)||\, dt$.
      iv. Thus we have a way to calculate the line integral of $f$ over $C$:

$$\int_C f(x, y, z)\, ds = \int_a^b f(\mathbf{r}(t))||\mathbf{r}'(t)||\, dt.$$

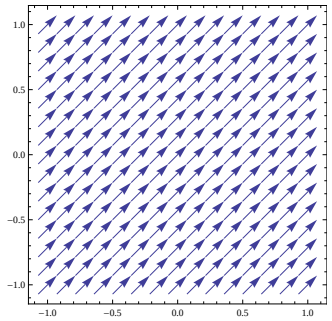
   c. Example: Find the mass of a spring in the shape of the circular helix $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$, $0 \le t \le 2\pi$, where the density at $(x, y, z)$ is given by $f(x, y, z) = 1 + z$.
      i. $\mathbf{r}'(t) = -\sin t\mathbf{i} + \cos t\mathbf{j} + \mathbf{k}$
      ii. $||\mathbf{r}'(t)|| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$.
      iii. Then $\displaystyle\int_C f(x, y, z)\, ds = \int_0^{2\pi} (1 + t)\sqrt{2}\, dt = \sqrt{2}\left[t + \frac{t^2}{2}\right]_0^{2\pi} = \sqrt{2}(2\pi + 2\pi^2) = 2\sqrt{2}\pi(1 + \pi)$.

II. The idea of a vector line integral
  a. Introduction
    i. Suppose we have a curve moving through a vector field, like so:




(Mathematica command: `VectorPlot[{x, y}, {x, -1, 1}, {y, -1, 1}]`.)

    ii. The line integral along the curve through the vector field measures how much the curve is going with, or against, the vector field overall.
    iii. This implies that the curve has a direction, or orientation.
    iv. Example: Work done by a force field in moving an object through space.
      1. If the object mostly moves with the force field then the work is positive.
      2. If the object mostly moves against the force field then the work is negative. (Some other entity is doing work against the force field.)
  b. Definition of the line integral
    i. Suppose we have some straight curve moving through a constant vector field, where $\mathbf{C}$ is a vector representing the curve:




      1. If the curve moves in the direction of the vector field then the line integral is just the length of the curve times the magnitude of the vector field $\mathbf{F}$, or $\|\mathbf{F}\|\|\mathbf{C}\|$.
        • You're picking up $\|\mathbf{F}\|$ over a distance $\|\mathbf{C}\|$.
      2. If the curve moves in the direction opposite $\mathbf{F}$ then the line integral is $-\|\mathbf{F}\|\|\mathbf{C}\|$.
      3. What if the curve moves at an angle to the vector field?
        • How much of the vector field is going with the curve?
        • We've got this situation:

- To find the "amount" of the vector field going with the curve we project the vector field vectors onto the curve.
- So the line integral is $\|\mathbf{F}\|\|\mathbf{C}\|\cos\theta$, where $\theta$ is the angle between the vector field vectors and $\mathbf{C}$.
- More generally, this can be expressed as $\mathbf{F}\cdot\mathbf{C}$.
- This derivation can also be done in terms of projections.

ii. What if the curve isn't straight, or the vector field isn't constant?

1. We chop the curve up into a bunch of tiny pieces so small that they are almost straight and so that the vector field is almost constant on each piece.

- On each of these little pieces we calculate how much of the vector field is going with the curve.
- Then we add up all the results from all the little pieces.
- If we let the length of the longest piece go to 0 then we get an exact value.

2. More formally
   - We represent the curve $C$ by some vector-valued function $\mathbf{r}(t)$, as that's the more general way to express curves.
   - The contribution from curve piece $i$ is $\mathbf{F}(\mathbf{r}_i)\cdot\Delta\mathbf{r}_i$, the vector field evaluated at $\mathbf{r}_i$ dotted with the vector describing piece $i$ of the curve.

- Thus the line integral over the curve $C$ is defined as

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \lim_{\|\Delta\mathbf{r}_i\|\to 0}\sum_{i=1}^{n}\mathbf{F}(\mathbf{r}_i)\cdot\Delta\mathbf{r}_i.$$

3. Caveats
   - The vector field $\mathbf{F}$ must be continuous over the curve $C$.
     - This means that all of the component functions of $\mathbf{F}$ must be continuous on $C$.
   - The curve $C$ must be piecewise smooth.
     - This means that $C$ must be made up of a finite number of smooth (no kinks) curves.

c. Conceptual examples: For different transparencies, ask if the line integral is positive, negative, or zero.

d. Interpretations of line integrals

i. Work

1. Work is force acting through a distance.
2. More specifically, the work done by a constant force $\mathbf{F}$ on an object moving in a straight line through a displacement $\mathbf{d}$ is $\mathbf{F}\cdot\mathbf{d}$.

3. If the force is variable, or the object doesn't move in a straight line, then we need the work integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

ii. Circulation

1. If the curve $C$ is closed then the line integral of $\mathbf{F}$ over $C$ is called the *circulation* of $\mathbf{F}$ around $C$.
2. So the circulation of $\mathbf{F}$ around $C$ measures the overall tendency of the vector field $\mathbf{F}$ to rotate along $C$.
3. Show with transparencies.

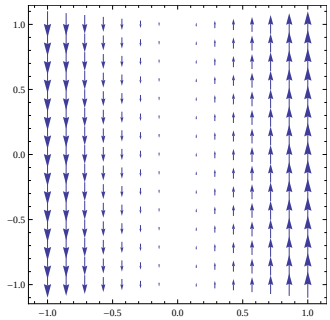
iii. Electric potential

1. Electric potential difference (commonly called voltage) between points $A$ and $B$ is defined as $\Delta V = -\int_A^B \mathbf{E} \cdot d\mathbf{r}$, where $\mathbf{E}$ is the electric field.
2. There is no voltage between the points in the following diagrams:

iv. Ampère's law also involves a line integral: $\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 I$.

1. It says that the line integral along a closed path through a magnetic field is equal to $\mu_0$ times the amount of current flowing through the loop of the path.
2. Here, $\mu_0$ is called the *permeability of free space*.
3. This is actually a special case of Stokes's Theorem.

e. Example: Compute the line integral of the vector field $\mathbf{F} = x\mathbf{j}$ along the line from (1,0) to (3,0).




i. The curve $C$ is orthogonal to the vector field $\mathbf{F}$.

ii. And thus $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$.

f. Properties of line integrals

i. Line integrals share a lot of the same properties of single-variable integrals.

ii. For a scalar constant $\lambda$, vector fields $\mathbf{F}$ and $\mathbf{G}$, and oriented curves $C, C_1, C_2$, we have

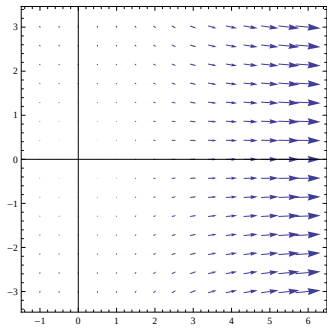
1. $\int_C \lambda\mathbf{F} \cdot d\mathbf{r} = \lambda \int_C \mathbf{F} \cdot d\mathbf{r}$.
2. $\int_C (\mathbf{F} + \mathbf{G}) \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r} + \int_C \mathbf{G} \cdot d\mathbf{r}$.
3. $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}$. (Illustrate this one.)
4. $\int_{C_1+C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_1} \mathbf{F} \cdot d\mathbf{r}$. (Illustrate this one.)

III. Computing line integrals

a. Basic idea

i. We've talked about what a line integral means and how to define one, and we've even calculated the values for a few line integrals, but how do you determine the value of a line integral in general?
ii. The basic idea is to convert the line integral to a definite integral in one variable.
iii. Then you can evaluate the integral using single-variable calculus.
iv. But how do you convert to a single integral?
    1. You work with the parametrization of the curve.
    2. Suppose we want to determine $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where the curve $C$ is given by $\mathbf{r}(t)$, $a \leq t \leq b$.
    3. We have $\dfrac{d\mathbf{r}(t)}{dt} = \mathbf{r}'(t)$.
    4. Thus $d\mathbf{r} = \mathbf{r}'(t)dt$.
    5. So we have $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F} \cdot \mathbf{r}'(t)\, dt$.
    6. Hidden in the notation, though, is the fact that we're evaluating $\mathbf{F}$ at $\mathbf{r}$, so this could be expressed as $\displaystyle\int_a^b \mathbf{F}(\mathbf{r}) \cdot \mathbf{r}'(t)\, dt$.
v. Comments
    1. The value of the line integral does not depend on the particular parameterization of the curve you pick.
        - This should make sense, as the line integral definition says nothing about parameterizations.
    2. This works regardless of how many dimensions you're in.

b. Examples
    i. Find the work done by the force $\mathbf{F}(x, y) = x^2\mathbf{i} - y\mathbf{j}$ over the curve $x = y^2$ from the point $(4, 2)$ to the point $(1, -1)$. (Problem 14.2 #19, Hass, Weir, and Thomas)
        1. Vector field and path:



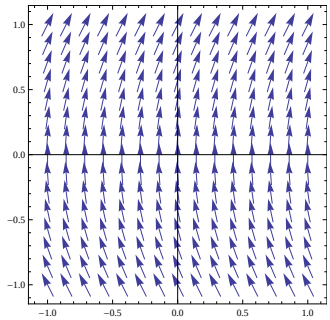

        2. Should the line integral be positive or negative? Negative.
        3. Parameterize the path.
            - Let $y = t$.
            - Then $x = t^2$.
            - Thus $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j}$, where $t$ runs from 2 to $-1$.
        4. We have $\mathbf{r}'(t) = 2t\mathbf{i} + \mathbf{j}$.
        5. Then $\mathbf{F}(\mathbf{r}(t)) = t^4\mathbf{i} - t\mathbf{j}$.
        6. So $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 2t^5 - t$.

7. Thus

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_2^{-1} (2t^5 - t)dt = \frac{1}{3}t^6 - \frac{1}{2}t^2\Big|_2^{-1} = \frac{1}{3} - \frac{1}{2} - \left(\frac{64}{3} - \frac{4}{2}\right) = -\frac{63}{3} + \frac{3}{2} = -21 + \frac{3}{2} = -19.5.$$

ii. Note the five steps in the process of evaluating the previous line integral:
   1. Parameterize the curve to obtain $\mathbf{r}(t)$.
   2. Differentiate to find $\mathbf{r}'(t)$.
   3. Substitute $\mathbf{r}(t)$ into $\mathbf{F}$ to obtain $\mathbf{F}(\mathbf{r}(t))$.
   4. Find $\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)$.
   5. Evaluate $\displaystyle\int_a^b \mathbf{F}(\mathbf{r})\cdot\mathbf{r}'(t)\,dt$.

iii. Compute $\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F} = y\mathbf{i} + 2\mathbf{j}$ for the following three curves from $(1,0)$ to $(0,1)$: The arc of a quarter circle, the straight line between the points, and the $L$-shaped path going through $(0,0)$.
   1. Vector field:




   2. Should all three line integrals give the same answer?
      - No.
      - Ask students to rank line integrals from greatest to smallest value.
   3. The quarter-circle from $(1,0)$ to $(0,1)$
      - Parameterization of curve: $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$, $0 \le t \le \pi/2$.
      - We have $\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t) = (\sin t\mathbf{i} + 2\mathbf{j})\cdot(-\sin t\mathbf{i} + \cos t\mathbf{j}) = -\sin^2 t + 2\cos t$.
      - Thus

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^{\pi/2} (-\sin^2 t + 2\cos t)\,dt = \int_0^{\pi/2} \left(-\frac{1-\cos 2t}{2} + 2\cos t\right)\,dt$$
$$= -\frac{t}{2} + \frac{\sin 2t}{4} + 2\sin t\Big|_0^{\pi/2} = -\frac{\pi}{4} - \frac{\sin\pi}{4} + 2\sin(\pi/2) - 0$$
$$= 2 - \frac{\pi}{4} \approx 1.215.$$

   4. The straight line from $(1,0)$ to $(0,1)$
      - Parameterization of curve: $\mathbf{r}(t) = (1-t)\mathbf{i} + t\mathbf{j}$, $0 \le t \le 1$.
      - We have $\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t) = (t\mathbf{i} + 2\mathbf{j})\cdot(-\mathbf{i} + \mathbf{j}) = -t + 2$.
      - Thus $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^1 (-t+2)\,dt = -\frac{1}{2}t^2 + 2t\Big|_0^1 = -\frac{1}{2} + 2 = 1\frac{1}{2}$.
   5. The $L$-shaped curve from $(1,0)$ to $(0,0)$ and then from $(0,0)$ to $(0,1)$
      - This will have to be done in two pieces.
      - First piece

- Parameterization of curve: $\mathbf{r}(t) = t\mathbf{i}$, $t$ goes from 1 to 0.
- $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 2\mathbf{j} \cdot \mathbf{i} = 0$.
- Thus $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^0 0\, dt = 0$.

• Second piece
- Parameterization of curve: $\mathbf{r}(t) = t\mathbf{j}$, $0 \le t \le 1$.
- $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = (t\mathbf{i} + 2\mathbf{j}) \cdot \mathbf{j} = 2$.
- Thus $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 2\, dt = 2t|_0^1 = 2$.

IV. Alternative notation for line integrals

a. $\displaystyle\int_C \mathbf{F} \cdot \mathbf{T}\, ds$ notation

i. Define $\mathbf{T}(t)$ to be the unit vector tangent to the graph of $\mathbf{r}(t)$ at $t$ and pointing in the direction that $\mathbf{r}(t)$ is pointing.

ii. Thus $d\mathbf{r}$ points in the direction of $\mathbf{T}$.

iii. This means that

$$\frac{d\mathbf{r}}{\|d\mathbf{r}\|} = \mathbf{T}.$$

iv. But $d\mathbf{r}$ is a piece of the underlying curve, so $\|d\mathbf{r}\| = ds$.

v. Thus we have $d\mathbf{r} = \mathbf{T}\, ds$.

vi. So we could express $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$ as $\int_C \mathbf{F} \cdot \mathbf{T}\, ds$.

vii. In fact, the text uses this notation in several places.

b. Differential notation

i. Another way to express $\Delta\mathbf{r}$ is as $\Delta x\mathbf{i} + \Delta y\mathbf{j}$:

ii. In differential form, this becomes $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j}$, or, in 3D, $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$.

iii. Thus if the vector field is given by $\mathbf{F}(x, y, z) = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ we have

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) = \int_C F_1 dx + F_2 dy + F_3 dz.$$

iv. Example: Evaluate $\int_C xy\, dx + z^2\, dy + (z - x)\, dz$ over the curve running from the top of the sphere of radius 2 centered at the origin straight down its side to the positive $y$-axis.

1. First, parameterize the curve.
   - The $x$ component is 0.
   - Projection onto $yz$-plane, graphically:

   - So one parameterization is $x = 0$, $y = 2\cos t$, $z = 2\sin t$, where $t$ ranges from $\pi/2$ down to 0.

2. Then we have

$$\int_{\pi/2}^{0} \big(0(2\cos t)(0) + 4\sin^2 t(-2\sin t) + (2\sin t - 0)(2\cos t)\big)\ dt$$
$$= \int_{\pi/2}^{0} \big(-8\sin^3 t + 4\sin t\cos t\big)\ dt = \int_{\pi/2}^{0} \big(-8\sin t(1-\cos^2 t) + 4\sin t\cos t\big)\ dt$$
$$= \int_{\pi/2}^{0} \big(-8\sin t + 8\sin t\cos^2 t + 4\sin t\cos t\big)\ dt = 8\cos t - \frac{8}{3}\cos^3 t + 2\sin^2 t\Big|_{\pi/2}^{0}$$
$$= 8 - \frac{8}{3} - 2 = 3\,\frac{1}{3}.$$

**Homework**: Section 16.2 #1, 2, 5, 8, 9, 15, 19, 22, 28, 35, 41, 42, 44, 52, 60.